

PGS CATALOG DATASETS

DATASET NAME : PGS000013

Polygenic Score (PGS) ID PGS000013 was released in the [PGS Catalog](#) on October 14, 2019. It predicts [coronary artery disease](#), which is mapped to the ontology term MONDO_0005010. The score is freely available to the academic community for research use, though parties interested in commercial applications must contact the Broad Office of Strategic Alliances and Partnering. The score, named GPS_CAD, was developed using the LDpred method with parameters set to $\rho = 0.001$ and an LD panel consisting of 503 1000G Europeans. It is built on the original genome build hg19 and includes 6,630,150 variants. The effect weight type is not reported. The score was originally published by [Khera AV et al. Nat Genet \(2018\)](#) and is cataloged under the publication ID [PGP000006](#).

Ancestry Distribution

The variant associations (GWAS) were sourced from a total of 187,599 individuals. The ancestry breakdown for this group is 75.3% European, 13.6% South Asian, 6% East Asian, 2.2% Hispanic or Latin American, 1.7% African, and 1.2% Greater Middle Eastern. The score development and training phase exclusively utilized 120,280 individuals of European ancestry (100%). Across 67 evaluation sample sets, the ancestry distribution is 47.8% European, 14.9% Multi-ancestry (including European), 13.4% Hispanic or Latin American, 9% African, 6% South Asian, 4.5% East Asian, 1.5% Additional Asian Ancestries, and 3% not reported.

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from [GCST003116](#) (Europe PMC: [26343387](#)). This included 141,217 European individuals, 25,557 South Asian individuals, 11,323 East Asian individuals, 4,095 Hispanic or Latin American individuals, and 3,139 African American or Afro-Caribbean individuals, all distributed across 41 cohorts. An additional 2,268 Greater Middle Eastern individuals were sourced from 40 cohorts. For the score's development and training, the model utilized 120,280 European individuals from the UKB (UK Biobank Phase 1) cohort. In this training set, coronary artery disease ascertainment was based on a composite of myocardial infarction or coronary revascularization.

Performance Metrics

For the Khera AV et al. (2018) publication, metric PPM000022 evaluated reported coronary artery disease in 288,978 European individuals (PSS000015), achieving an AUROC of 0.81, adjusted for age, sex, Ancestry PC 1-4, and genotyping chip.

Wünnemann F et al. (2019) reported several metrics for prevalent and recurrent coronary artery disease. Metric PPM000030 evaluated 1,964 European individuals (PSS000021), showing an Odds Ratio (OR) of 1.64 and an AUROC of 0.72. Metric PPM000031 assessed

3,309 European individuals (PSS000022), reporting an OR of 1.55 and an AUROC of 0.89. Metric PPM000032 examined 5,762 European individuals (PSS000019) with an OR of 1.69 and an AUROC of 0.84. Furthermore, metric PPM000033 looked at recurrent events in 3,195 European individuals (PSS000020), yielding an OR of 1.13. All Wünnemann metrics adjusted for age, sex, and the first four genetic PCs, with PPM000032 also accounting for the cohort recruitment center.

Homburger JR et al. (2019) evaluated personal history of coronary artery disease in 11,010 European individuals (PSS000219). Metric PPM000383 reported an OR of 1.589 and an AUROC of 0.86 adjusted for age and sex, while metric PPM000387 reported an AUROC of 0.6 with no covariates listed.

Khera AV et al. (2019) focused on early-onset myocardial infarction (age $\leq$ 55 years), adjusting for 4 genetic PCs across various ancestries. Metric PPM000399 evaluated 3,081 European individuals (PSS000230) with an OR of 2.06. Metric PPM000400 assessed 1,298 African individuals (PSS000228) with an OR of 1.46. Metric PPM000401 evaluated 919 Hispanic or Latin American individuals (PSS000229) with an OR of 1.56. Metric PPM000402 evaluated 544 individuals of additional Asian ancestries (PSS000227) with an OR of 2.16.

Dikilitas O et al. (2020) analyzed incident and prevalent coronary heart disease using age-as-time-scale Cox regression, adjusting for sex, eMERGE site, and the first five ancestry PCs. For incident disease, metric PPM000590 evaluated 39,758 European individuals (PSS000334) with a Hazard Ratio (HR) of 1.5 and a C-index of 0.719. Metric PPM000593 assessed 7,070 African individuals (PSS000332) with an HR of 1.19 and a C-index of 0.656. Metric PPM000596 evaluated 2,194 Hispanic or Latin American individuals (PSS000336) with an HR of 1.16 and a C-index of 0.659. For incident and prevalent disease combined, metric PPM000599 evaluated 45,645 European individuals (PSS000333), reporting an OR of 1.66 and an AUROC of 0.77, additionally adjusting for age at the first EHR record and EHR duration.

Evaluated Samples

The UKB Phase 2 cohort (PSS000015) included 288,978 European individuals, identifying CAD through a composite of myocardial infarction or revascularization. The CARTaGENE cohort (PSS000019) consisted of 5,762 French Canadian European individuals (41.29% male), evaluating prevalent CAD.

The MHI cohort contributed multiple sample sets: PSS000020 evaluated recurrent CAD events in two phases involving 862 and 2,333 French Canadian European individuals. Prevalent CAD was assessed in PSS000021 (Phase 1) with 1,964 French Canadian European individuals (72.7% male) and in PSS000022 (Phase 2) with 3,309 French Canadian European individuals (72.38% male).

The CG cohort (PSS000219) contained 11,010 European individuals (17.1% male) whose phenotype information was self-reported via a Color Genomics multi-gene panel test health history. Cases from the VIRGO cohort and controls from the MESA cohort were combined to evaluate multiple ancestries: PSS000227 included 544 Asian individuals, PSS000228 included 1,298 African American or Afro-Caribbean individuals, PSS000229 included 919

Hispanic or Latin American individuals, and PSS000230 included 3,081 European individuals.

Data from 7 cohorts was used to define CHD in African American or Afro-Caribbean individuals (mean age 43.6, 31% male) for PSS000331 (7,597 individuals) and PSS000332 (7,070 individuals). Similarly, data from 11 cohorts assessed CHD in European individuals (mean age 49.0, 44.6% male) for PSS000333 (45,645 individuals) and PSS000334 (39,758 individuals).

DATASET NAME : PGS000018

Polygenic Score (PGS) ID PGS000018 was released in the [PGS Catalog](#) on October 14, 2019. It predicts coronary artery disease, which maps to the ontology term [coronary artery disorder](#) ([MONDO_0005010](#)). The score, named metaGRS_CAD, was developed using the metaGRS method with specific log(HR) mixing weights parameters: GRS46K=0.1278, FDR202=0.2359, and 1000Genomes=0.2400. It is based on the original genome build hg19 and consists of 1,745,179 variants, though the effect weight type is not reported (NR). The score is documented under the publication ID [PGP000007](#) and was originally cited in [Inouye M et al. J Am Coll Cardiol \(2018\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 382,026 individuals. The ancestry breakdown for this GWAS group is 50.9% Multi-ancestry (including European), 37% European, 6.7% South Asian, 3% East Asian, 1.1% Hispanic or Latin American, 0.8% African, and 0.6% Greater Middle Eastern. During the score development and training phase, 3,000 individuals were used, consisting of 100% Multi-ancestry (including European) individuals. Across 36 evaluation sample sets, the ancestry distribution is reported as 63.9% European, 16.7% Hispanic or Latin American, 8.3% African, 8.3% Multi-ancestry (including European), and 2.8% Greater Middle Eastern.

Development Samples

The Genome-Wide Association Studies (GWAS) utilized data from multiple sources. According to [Europe PMC: 23202125](#), 194,427 European and South Asian individuals were sourced across 34 cohorts. Additional data from [GWAS Catalog: GCST003116](#) and [Europe PMC: 26343387](#) included 141,217 European individuals, 25,557 South Asian individuals, 11,323 East Asian individuals, 4,095 Hispanic or Latin American individuals, and 3,139 African American or Afro-Caribbean individuals across 41 cohorts. Furthermore, 2,268 Greater Middle Eastern individuals were drawn from 40 cohorts in the same studies. For the score's development and training, 3,000 individuals of European and Not Reported (NR) ancestries were sourced from the UKB cohort, consisting of ~95% European ancestry and <5% non-European ancestry. In this training set, CAD was defined as fatal or nonfatal myocardial infarction cases.

Performance Metrics

Inouye M et al. (2018) reported performance metrics for incident coronary artery disease under metric PPM000027, evaluating 482,629 multi-ancestry individuals ([PSS000018](#)). Using age-as-time-scale Cox regression adjusted for sex, genetic PCs 1-10, and genotyping array, the model achieved a Hazard Ratio (HR) of 1.706, an AUROC of 0.79, a C-index of 0.623, and an AUPRC of 0.161.

Wünnemann F et al. (2019) provided multiple metrics evaluating prevalent and recurrent CAD in European individuals, all adjusting for age, sex, and the first four genetic PCs. For prevalent CAD, metric PPM000034 assessed 1,964 individuals ([PSS000021](#)) with an OR of 1.74 and an AUROC of 0.72. Metric PPM000035 evaluated 3,309 individuals ([PSS000022](#)) with an OR of 1.6 and an AUROC of 0.89. Metric PPM000036 examined 5,762 individuals ([PSS000019](#)), additionally adjusting for cohort recruitment centre, yielding an OR of 1.75 and an AUROC of 0.84. For recurrent CAD events, metric PPM000037 evaluated 3,195 individuals ([PSS000020](#)) with an OR of 1.17.

Timmerman N et al. (2020) evaluated a European cohort of 1,319 individuals ([PSS000287](#)) across several traits. For secondary cardiovascular events (PPM000498), they found an HR of 1.15, adjusting for age, sex, diabetes, BMI, smoking, hypercholesterolemia, array, and 4 genetic PCs. Additional metrics adjusted for age, sex, surgery year, type of cerebrovascular symptoms, array, and 4 genetic PCs, focusing on comparisons of the top 20% versus the rest. These included moderate/heavy calcifications (PPM000499; OR 1.001), moderate/heavy collagen (PPM000500; OR 1.091), presence of lipid core >10% (PPM000501; OR 1.591), presence of intraplaque hemorrhage (IPH) (PPM000502; OR 1.112), moderate/heavy smooth muscle cells (PPM000503; OR 0.908), and moderate/heavy macrophages (PPM000504; OR 1.49). Beta values were also reported for the number of macrophages (PPM000505; Beta 0.55), number of smooth muscle cells (PPM000506; Beta -0.056), and microvessels (PPM000507; Beta 0.072).

Evaluated Samples

The UKB cohort ([PSS000018](#)) included 482,629 individuals (~95% European, 45.6% male), defining CAD as fatal or nonfatal myocardial infarction. The CARTaGENE cohort ([PSS000019](#)) involved 5,762 French Canadian European individuals (41.29% male) for evaluating prevalent CAD.

The MHI cohort was split into multiple groups for French Canadian Europeans. For recurrent CAD events, [PSS000020](#) was evaluated in Phase 1 (862 individuals) and Phase 2 (2,333 individuals) with a median follow-up of 3.9 years. Prevalent CAD was assessed in Phase 1 ([PSS000021](#)) with 1,964 individuals (72.7% male) and in Phase 2 ([PSS000022](#)) with 3,309 individuals (72.38% male).

The AEGS1 cohort ([PSS000287](#)) studied 1,319 Dutch European individuals (69.3% male, mean age 68.8) to track secondary cardiovascular events over a mean follow-up of 3.0 years. Data from 7 cohorts was used to define CHD in African American or Afro-Caribbean individuals (mean age 43.6, 31.0% male) for [PSS000331](#) (7,597 individuals) and [PSS000332](#) (7,070 individuals), both featuring a median follow-up of 9.2 years. Similarly, 11

cohorts provided data for European individuals (mean age 49.0, 44.6% male) for [PSS000333](#) (45,645 individuals) and [PSS000334](#) (39,758 individuals) with a median follow-up of 11.7 years.

Hispanic or Latin American individuals (mean age 41.1, 36.2% male) were evaluated using 8 cohorts for [PSS000335](#) (2,493 individuals) and [PSS000336](#) (2,194 individuals) with a median follow-up of 10.4 years. The INTERVAL cohort ([PSS000868](#)) included 3,087 European individuals (51.0% male, median age 44.0), utilizing CALIBER rule-based phenotyping with a median follow-up of 6.9 years. Finally, the GerMIFS and LURIC cohorts ([PSS000929](#)) included 5,581 European individuals, defining CAD as myocardial infarction before age 60.

DATASET NAME : PGS000012

Polygenic Score (PGS) ID PGS000012 was released in the [PGS Catalog](#) on October 14, 2019. It is designed to predict coronary artery disease, which maps to the [coronary artery disorder](#) ontology term ([MONDO_0005010](#)). The score, named GRS49K, was constructed using the LD thinning development method with an r^2 threshold parameter of 0.7. It was built on the original genome build hg19 and includes 49,310 variants. The effect weight type is not reported (NR). The score's catalog publication ID is [PGP000005](#), derived from the original citation by [Abraham G et al. Eur Heart J \(2016\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 194,427 individuals, categorized entirely (100%) under a Multi-ancestry (including European) classification. During the score development and training phase, 5,883 individuals were utilized, consisting of 100% European ancestry. Across the 5 evaluated sample sets, the ancestry distribution is reported as 80% European and 20% Multi-ancestry (including European).

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from [Europe PMC: 23202125](#), encompassing 194,427 European and South Asian individuals distributed across 33 cohorts. For the score's development and training, data was drawn from two GWAS Catalog subsets. [GCST000340](#) provided 1,019 European individuals from the MIGen cohort (specifically the Harps subset), where cases were defined as early-onset MI (men <50 years old, women <60 years old). Additionally, [GCST000045](#) contributed 4,864 European individuals from the WTCCC cohort, where CAD was defined as a validated history of either myocardial infarction or coronary revascularization.

Performance Metrics

Abraham G et al. (2016) reported performance metrics for incident coronary artery disease. Metric PPM000018 evaluated 12,676 European individuals ([PSS000012](#)) using only the 42,364 SNPs available in the FINRISK dataset. This model achieved both a Hazard Ratio (HR) and Odds Ratio (OR) of 1.74, adjusting for sex, sub-cohort, location (east/west), and 5 genetic PCs. Metric PPM000020 evaluated 3,406 European individuals ([PSS000011](#))

utilizing the 46,773 SNPs available in FHS, yielding an HR and OR of 1.28 with adjustments for sex, sub-cohort, and 5 genetic PCs.

Inouye M et al. (2018) contributed metric PPM000028 for incident coronary artery disease in 482,629 multi-ancestry individuals ([PSS000018](#)). The evaluation used the GRS46K variant and achieved an HR of 1.524, adjusting for sex, genetic PCs (1-10), and genotyping array.

Liou L et al. (2021) examined incident coronary artery disease specifically in individuals with breast cancer under metric PPM005158. Evaluating 8,946 European individuals ([PSS003596](#)), the model reported an HR of 1.31 while adjusting for age at diagnosis, genotype array, PCs(1-8), body mass index, smoking, and sociodemographic variables.

Martikainen P et al. (2021) provided multiple metrics predicting incident coronary heart disease hospitalization or death in 26,203 European individuals ([PSS011680](#)), tracking the Hazard Ratio of the top 25% versus the bottom 25%. Metric PPM021300 achieved an HR of 2.26, adjusting for age as a timescale, sex, region of residence, calendar year, study batch, and PCs(1-10). Metric PPM021301 yielded an HR of 2.27 by adding education to the previous covariates. Metric PPM021302 reported an HR of 2.12 when further incorporating smoking and alcohol covariates.

Evaluated Samples

The FHS cohort (including FHS Original and FHS Offspring) tracked incident CHD events before age 75 in 3,406 European individuals (45.0% male) under sample set [PSS000011](#).

The FINRISK cohorts contributed to two distinct sample sets. [PSS000012](#) utilized data from FR92, FR97, and FR02 to evaluate CHD in 12,676 European (Finnish) individuals (46.0% male). A broader evaluation in [PSS011680](#) monitored 26,203 European (Finnish) individuals (46.74% male, mean age 48.0) who were free of CHD at baseline, tracking incident cases over a mean follow-up period of 13.8 years.

The UKB cohort ([PSS000018](#)) consisted of 482,629 individuals (45.6% male), comprising ~95% European ancestry and <5% non-European ancestry. In this dataset, CAD was defined as fatal or nonfatal myocardial infarction cases.

Finally, the SEARCH cohort ([PSS003596](#)) uniquely focused on 8,946 European individuals (0.0% male), all of whom had a breast cancer diagnosis, to document incident coronary artery disease.

DATASET NAME : PGS000116

Polygenic Score (PGS) ID PGS000116 was released in the [PGS Catalog](#) on March 27, 2020. It predicts coronary artery disease, which maps to the [coronary artery disorder](#) ontology term ([MONDO_0005010](#)). The score is freely available under the CC0 1.0 Universal Public Domain Dedication. The score, named CAD_EJ2020, was developed using the lassosum method with parameters set to INFO score>0.999; N=1,037,385 variants; s=0.5; and lambda=0.00428. It is built on the original genome build GRCh37 and consists of

40,079 variants. The effect weight type is not reported (NR). The score's catalog publication ID is [PGP000054](#), corresponding to the original citation by [Elliott J et al. JAMA \(2020\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 187,599 individuals. The ancestry breakdown for this GWAS group is 75.3% European, 13.6% South Asian, 6% East Asian, 2.2% Hispanic or Latin American, 1.7% African, and 1.2% Greater Middle Eastern. During the score development and training phase, 80,103 individuals were used, consisting entirely (100%) of Multi-ancestry (including European) individuals. Across 12 evaluation sample sets, the ancestry distribution is reported as 83.3% Multi-ancestry (including European) and 16.7% European.

Development Samples

The Genome-Wide Association Studies (GWAS) utilized data from [GWAS Catalog: GCST003116](#) and [Europe PMC: 26343387](#). This included 141,217 European individuals, 25,557 South Asian individuals, 11,323 East Asian individuals, 4,095 Hispanic or Latin American individuals, and 3,139 African American or Afro-Caribbean individuals distributed across 41 cohorts. An additional 2,268 Greater Middle Eastern (Middle Eastern, North African, or Persian) individuals were drawn from 40 cohorts. For the score's development and training, 80,103 individuals (68.24% male, mean age 60.0) from the UKB cohort were used as a tuning set. This training group consisted of 95% European, 1.4% Indian, 1.1% other ancestries, 0.5% Pakistani, 0.4% other Asian, and 0.2% Chinese ancestries. In this group, coronary artery disease was defined as myocardial infarction and its related sequelae.

Performance Metrics

Elliott J et al. (2020) reported multiple performance metrics predicting incident coronary artery disease across various multi-ancestry populations using data from the UKB cohort.

For the overall sample of 352,660 individuals ([PSS000399](#)), the model yielded a Hazard Ratio (HR) of 1.32 and a C-index of 0.61 with no covariates (PPM000806). The C-index improved to 0.76 when adjusting for age and sex (PPM000807), and to 0.78 when incorporating pooled cohort equations (PPM000808).

When evaluating incident coronary artery disease in individuals over age 55 across 204,675 individuals ([PSS000387](#)), the C-index was 0.6 without covariates (PPM000809), 0.71 adjusting for age and sex (PPM000810), and 0.74 adjusting for pooled cohort equations (PPM000811). Conversely, for 147,985 individuals under age 55 ([PSS000383](#)), the C-index was 0.64 without covariates (PPM000812), 0.76 with age and sex adjustments (PPM000813), and 0.80 using pooled cohort equations (PPM000814).

Gender-specific metrics were also evaluated. For 147,363 males ([PSS000391](#)), the C-index was 0.61 without covariates (PPM000815), 0.68 with age and sex (PPM000816), and 0.71 with pooled cohort equations (PPM000817). For 205,297 females ([PSS000395](#)), the C-index

was 0.61 without covariates (PPM000818), 0.71 adjusting for age and sex (PPM000819), and 0.76 using pooled cohort equations (PPM000820).

Evaluated Samples

The UKB cohort provided several evaluated sample sets with a mean follow-up time of 8.01 years, divided into PCE Prospective Cohorts (primarily European, African unspecified, and Not Reported; 98.3% White European and 1.7% Black/Other) and QRISK3 Prospective Cohorts (more diverse multi-ancestry groups).

For the PCE Prospective Cohorts, evaluations included 147,985 individuals aged 40-55 ([PSS000383](#)), 204,675 individuals aged 55-69 ([PSS000387](#)), 147,363 males with a mean age of 55.79 ([PSS000391](#)), 205,297 females with a mean age of 56.0 ([PSS000395](#)), and a combined group of 352,660 individuals across the 40-69 age range ([PSS000399](#)).

For the QRISK3 Prospective Cohorts, which included additional South Asian, East Asian, and Afro-Caribbean ancestries, evaluations included 147,110 individuals aged 40-55 ([PSS000385](#)), 203,620 individuals aged 55-69 ([PSS000389](#)), 146,573 males with a mean age of 55.8 ([PSS000393](#)), 204,157 females with a mean age of 56.0 ([PSS000397](#)), and a combined set of 350,730 individuals across the 40-69 age range ([PSS000401](#)).

Additionally, two sample sets from the CoLaus cohort were evaluated for CAD (non-fatal MI, death from CAD, or coronary revascularization) in European individuals. [PSS010121](#) assessed 3,383 individuals (45.0% male, mean age 52.3) with statin therapy at baseline excluded, while [PSS010122](#) assessed 4,218 individuals (47.0% male, mean age 53.4) over a median follow-up of 10.6 to 14.6 years.

DATASER NAME : PGS000117

Polygenic Score (PGS) ID PGS000117 was released in the [PGS Catalog](#) on March 27, 2020. It predicts cardiovascular disease, which maps to the [cardiovascular disorder](#) ontology term ([MONDO_0004995](#)). The score is freely available under the CC0 1.0 Universal Public Domain Dedication. Named CVD_EJ2020, the score was developed using the lassosum method with the parameters set to INFO score>0.999; N=1,037,385 variants; s=0.5; and lambda=0.00429. It is built on the original genome build GRCh37 and includes 297,862 variants, though the effect weight type is not reported (NR). The score's catalog publication ID is [PGP000054](#), matching the original citation by [Elliott J et al. JAMA \(2020\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 187,599 individuals. The ancestry breakdown for this GWAS group is 75.3% European, 13.6% South Asian, 6% East Asian, 2.2% Hispanic or Latin American, 1.7% African, and 1.2% Greater Middle Eastern. During the score development and training phase, 80,103 individuals were utilized, consisting entirely (100%) of Multi-ancestry (including European) individuals. Across 12

evaluation sample sets, the ancestry distribution is reported as 83.3% Multi-ancestry (including European) and 16.7% European.

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from [GWAS Catalog: GCST003116](#) and [Europe PMC: 26343387](#). This included 141,217 European individuals, 25,557 South Asian individuals, 11,323 East Asian individuals, 4,095 Hispanic or Latin American individuals, and 3,139 African American or Afro-Caribbean individuals, each distributed across 41 cohorts. An additional 2,268 Greater Middle Eastern (Middle Eastern, North African, or Persian) individuals were drawn from 40 cohorts. For the score's development and training, a case-control tuning set of 80,103 individuals (68.24% male, mean age 60.0) from the UKB cohort was used. This multi-ancestry training group consisted of 95% European, 1.4% Indian, 1.1% other ancestries, 0.5% Pakistani, 0.4% other Asian, and 0.2% Chinese ancestries. In this group, cardiovascular disease was defined as coronary artery disease, additionally including angina and non-fatal events.

Performance Metrics

Elliott J et al. (2020) reported multiple performance metrics predicting incident cardiovascular disease across various multi-ancestry populations using data from the UKB cohort.

For the overall sample of 352,660 individuals ([PSS000400](#)), the model yielded a C-index of 0.56 with no covariates (PPM000821). The C-index improved to 0.71 when adjusting for age and sex (PPM000822), and to 0.75 when incorporating pooled cohort equations (PPM000823).

When evaluating incident cardiovascular disease in 204,675 individuals categorized under age 55 ([PSS000388](#)), the C-index was 0.56 without covariates (PPM000824), 0.66 adjusting for age and sex (PPM000825), and 0.70 adjusting for pooled cohort equations (PPM000826). Conversely, for 147,985 individuals categorized over age 55 ([PSS000384](#)), the C-index was 0.59 without covariates (PPM000827), 0.70 with age and sex adjustments (PPM000828), and 0.77 using pooled cohort equations (PPM000829).

Gender-specific metrics were also evaluated. For 147,363 males ([PSS000392](#)), the C-index was 0.57 without covariates (PPM000830), 0.67 with age and sex (PPM000831), and 0.71 with pooled cohort equations (PPM000832). For 205,297 females ([PSS000396](#)), the C-index was 0.56 without covariates (PPM000833), 0.69 adjusting for age and sex (PPM000834), and 0.74 using pooled cohort equations (PPM000835).

Evaluated Samples

The UKB cohort provided several evaluated sample sets with a mean follow-up time of 8.01 years, divided into PCE Prospective Cohorts (primarily European, African unspecified, and Not Reported; 98.3% White European and 1.7% Black/Other) and QRISK3 Prospective Cohorts (more diverse multi-ancestry groups).

For the PCE Prospective Cohorts, evaluations included 147,985 individuals aged 40-55 ([PSS000384](#)), 204,675 individuals aged 55-69 ([PSS000388](#)), 147,363 males with a mean age of 55.79 ([PSS000392](#)), 205,297 females with a mean age of 56.0 ([PSS000396](#)), and a combined group of 352,660 individuals across the 40-69 age range ([PSS000400](#)). All these groups were noted to have lipid-lowering treatment performed.

For the QRISK3 Prospective Cohort/Testing Sets, which included additional South Asian, East Asian, and Afro-Caribbean ancestries, evaluations included 147,110 individuals aged 40-55 ([PSS000386](#)), 203,620 individuals aged 55-69 ([PSS000390](#)), 146,573 males with a mean age of 55.8 ([PSS000394](#)), 204,157 females with a mean age of 56.0 ([PSS000398](#)), and a combined set of 350,730 individuals across the 40-69 age range ([PSS000402](#)).

Additionally, two sample sets from the CoLaus cohort were evaluated for Atherosclerotic cardiovascular disease (ASCVD), comprising non-fatal acute myocardial infarction and death, in European individuals. [PSS010119](#) assessed 3,383 individuals (45.0% male, mean age 52.3) with a median follow-up of 10.7 to 14.6 years, with participants on statin therapy at baseline excluded. [PSS010120](#) assessed 4,218 individuals (47.0% male, mean age 53.4) over a median follow-up of 10.6 to 14.6 years.

DATASET NAME: PGS003356

Polygenic Score (PGS) ID PGS003356 was released in the [PGS Catalog](#) on December 20, 2022. It predicts coronary artery disease, which maps to the [coronary artery disorder](#) ontology term (MONDO_0005010). The score, named 1MH_CAD_PRS_2022_Ldpred, was developed using the LDpred method with parameters set to an LD panel consisting of 503 European ancestry individuals from the 1000 Genomes project (v.1.0). It is built on the original genome build GRCh37 and includes 2,324,683 variants, utilizing an effect weight type of beta. The score's catalog publication ID is PGP000409, corresponding to the original citation by [Aragam KG et al. Nat Genet \(2022\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 1,165,690 individuals. The ancestry breakdown for this GWAS group is entirely (100%) Multi-ancestry (including European). Across the 4 evaluated sample sets, the ancestry distribution is split into 50% European, 25% African, and 25% Not Reported.

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from [GWAS Catalog: GCST90132314](#) and [Europe PMC: 36474045](#). This dataset encompassed a total of 1,165,690 individuals of European and Not Reported (NR) ancestries, distributed across 12 distinct cohorts.

Performance Metrics

Aragam KG et al. (2022) reported metrics evaluating both incident and recurrent coronary artery disease. For incident coronary artery disease, metric PPM016208 evaluated 5,685 individuals of unreported ancestry (PSS010060), yielding a Hazard Ratio (HR) of 1.61, adjusted for age, sex, and ancestry (PCs 1-5). Metric PPM016209 evaluated the same cohort and achieved an HR of 1.54 when additionally adjusting for established CAD risk factors such as total and HDL cholesterol alongside age, sex, and ancestry. For recurrent coronary artery disease, metric PPM016211 evaluated 14,298 European individuals (PSS010059), yielding an HR of 1.2 with adjustments for age, sex, and ancestry (PCs 1-5).

Guarisch-Sousa R et al. (2023) provided additional metrics evaluating raised coronary lesions. Metric PPM020882 assessed 564 European individuals (PSS011442) with an Odds Ratio (OR) of 1.44 and no reported covariates. Metric PPM020897 evaluated 504 African individuals (PSS011441), yielding an OR of 1.0, also without reported covariates.

Evaluated Samples

Four distinct sample sets were evaluated to generate the performance metrics. The FOURIER cohort ([PSS010059](#)) tracked 14,298 European individuals over a median follow-up time of 2.3 years. The MDC cohort ([PSS010060](#)) included 5,685 individuals (41.0% male) of unreported ancestry. The PDAY cohort contributed two specific sample sets based on ancestry: [PSS011441](#) evaluated 504 African (unspecified) individuals (82.0% male) with a mean age of 27.5 years, while [PSS011442](#) evaluated 564 European individuals (77.0% male) with a mean age of 26.7 years.

DATASET NAME: PGS003725

Polygenic Score (PGS) ID PGS003725 was released in the [PGS Catalog](#) on July 5, 2023. It predicts coronary artery disease, which maps to the [coronary artery disorder](#) ontology term ([MONDO 0005010](#)). The score, named GPS_Mult, was developed using the LDpred2 method with varying p-value threshold parameters (e.g., 0.0001, 0.00018, 0.00032, 0.001, up to 0.18 and beyond). It is built on the original genome build hg19 and includes 1,296,172 variants. The effect weight type utilized for this score is beta. The score is officially documented under the catalog publication ID [PGP000466](#) and correlates with the citation by [Patel AP et al. Nat Med \(2023\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a massive sample size of 1,443,386 individuals, categorized entirely (100%) as Multi-ancestry (including European). During the score development and training phase, 116,649 individuals were utilized, consisting of 100% European ancestry. Across the 12 evaluated sample sets, the ancestry distribution is diverse: 33.3% Hispanic or Latin American, 16.7% African, 16.7% European, 16.7% South Asian, 8.3% East Asian, and 8.3% Multi-ancestry (including European).

Development Samples

The Genome-Wide Association Studies (GWAS) utilized data from 1,443,386 individuals distributed across 11 cohorts. The ancestries represented in this GWAS group include African unspecified, East Asian, European, Hispanic or Latin American, and South Asian. For the score's development and training, data was sourced from the UKB cohort, specifically using the first released tranche of genotypes from UKBB (Europe PMC: [30305743](#)). This training set consisted of 116,649 British European individuals (47.5% male) with a mean age of 57.5 years. Phenotypes in this group were defined using various ICD 10 codes for myocardial infarction and related conditions (e.g., I21, I22, I23).

Performance Metrics

Patel AP et al. (2023) reported numerous performance metrics for incident coronary artery disease, uniformly adjusting for age, sex, and the first ten principal components of genetic ancestry. Metric PPM018419 evaluated 308,264 European individuals ([PSS010960](#)), achieving a Hazard Ratio (HR) of 1.75 and an Odds Ratio (OR) of 2.14. Metric PPM018420 evaluated 7,281 African individuals ([PSS010961](#)) with an HR of 1.25 and OR of 1.39. Metric PPM018421 evaluated 1,464 East Asian individuals ([PSS010962](#)) with an HR of 1.72 and OR of 2.14. Metric PPM018422 evaluated 8,982 South Asian individuals ([PSS010963](#)) yielding an HR of 1.62 and OR of 2.02. Additionally, pure Odds Ratios were reported across different cohorts: PPM018423 for 33,096 African individuals ([PSS010964](#)) yielded an OR of 1.25; PPM018424 for 124,467 European individuals ([PSS010965](#)) yielded an OR of 1.72; PPM018425 for 16,433 Hispanic or Latin American individuals ([PSS010966](#)) yielded an OR of 1.61; and PPM018426 for 16,874 South Asian individuals ([PSS010967](#)) yielded an OR of 1.83.

Zou RS et al. (2024) reported metrics using a multi-ancestry cohort of 663 individuals ([PSS012103](#)), adjusting for age, sex, and the first 10 principal components of ancestry. They evaluated plaque presence (PPM023039) with an OR of 1.42, stenosis over 50% (PPM023040) with an OR of 2.39, and visible non-calcified or vulnerable plaque (PPM023041) with an OR of 1.45.

Liu T et al. (2026) provided metrics for coronary heart disease in Hispanic/Latin American individuals, adjusting for age, sex, PC1-PC7, baseline waist-to-hip ratio, blood pressure, and smoking status. Metric PPM036571 evaluated 133,207 individuals ([PSS012313](#)) with an OR of 1.24 and AUROC of 0.749. Metric PPM036579 evaluated 43,338 individuals ([PSS012314](#)) with an OR of 1.33 and AUROC of 0.729. Metric PPM036587 evaluated 89,869 individuals ([PSS012315](#)) with an OR of 1.17 and AUROC of 0.747.

Evaluated Samples

The UKB cohort provided several sample sets for evaluating coronary artery disease, uniformly reflecting a mean participant follow-up time of 12.0 years. These included [PSS010960](#) (308,264 European individuals, 45.6% male, mean age 57.3), [PSS010961](#) (7,281 African unspecified individuals, 43.5% male, mean age 52.4), [PSS010962](#) (1,464 East Asian individuals, 37.2% male, mean age 53.0), and [PSS010963](#) (8,982 South Asian individuals, 54.1% male, mean age 53.8). The European UKB set explicitly excluded the first released tranche of genotypes used in the training phase.

The MVP cohort was used to evaluate multiple ancestries, excluding individuals from a previously published GWAS (GCST90132302). These sets included [PSS010964](#) (33,096 African American/Afro-Caribbean individuals), [PSS010965](#) (124,467 European individuals), and [PSS010966](#) (16,433 Hispanic or Latin American individuals).

The G&H cohort ([PSS010967](#)) evaluated 16,874 South Asian (Bangladeshi, Pakistani) individuals, excluding participants from published GWAS GCST90140952. The REPRIEVE cohort ([PSS012103](#)) evaluated a combined multi-ancestry sample composed of 277 African unspecified, 379 European, 6 East Asian, and 1 South Asian individuals.

Finally, the MCPS cohort provided evaluations of Hispanic or Latin American (Mexican) individuals. [PSS012313](#) evaluated a mixed-gender group of 133,207 individuals (33.0% male, mean age 51.2) over a median follow-up of 20.3 years. The cohort was also separated by gender, with [PSS012314](#) assessing 43,338 males (mean age 52.0) and [PSS012315](#) assessing 89,869 females (mean age 50.8).

DATASET NAME: PGS003726

Polygenic Score (PGS) ID PGS003726 was released in the [PGS Catalog](#) on July 5, 2023. It predicts coronary artery disease, which maps to the [coronary artery disorder](#) ontology term (MONDO_0005010). The score, named GPS_CADANC, was developed using the LDpred2 method with parameters set to $p = [0.0018, 0.0056, 0.0032, 0.018, 0.01, 1]$, h^2 scale = $[0.7, 1, 1.4]$, and Sparse LD = $[T, F]$. It is built on the original genome build hg19 and includes 1,296,172 variants. The effect weight type utilized for this score is beta. The score is officially documented under the catalog publication ID [PGP000466](#) and corresponds with the citation by [Patel AP et al. Nat Med \(2023\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a massive sample size of 1,443,386 individuals, categorized entirely (100%) as Multi-ancestry (including European). During the score development and training phase, 116,649 individuals were utilized, consisting of 100% European ancestry. Across the 4 evaluated sample sets, the ancestry distribution is evenly split into 25% African, 25% East Asian, 25% European, and 25% South Asian.

Development Samples

The Genome-Wide Association Studies (GWAS) utilized data from 1,443,386 individuals distributed across the BBJ, CARDIoGRAMplusC4D, FinnGen, G&H, and MVP cohorts. The ancestries represented in this GWAS group include African unspecified, East Asian, European, Hispanic or Latin American, and South Asian. For the score's development and training, data was sourced from the UKB cohort, specifically using the first released tranche of genotypes from UKBB. This training set consisted of 116,649 European (British) individuals (47.5% male) with a mean age of 57.5 years. Phenotypes in this group were

defined using various ICD 10 codes for myocardial infarction and related conditions (e.g., I21, I22, I23).

Performance Metrics

Patel AP et al. (2023) reported several performance metrics for predicting coronary artery disease, uniformly adjusting for age, sex, and the first ten principal components of genetic ancestry. Metric PPM018427 evaluated 308,264 European individuals ([PSS010960](#)), achieving a Hazard Ratio (HR) of 1.73 (95% CI: 1.69-1.76). Metric PPM018428 evaluated 7,281 African ancestry individuals ([PSS010961](#)) with an HR of 1.18 (95% CI: 1.01-1.37). Metric PPM018429 evaluated 1,464 East Asian ancestry individuals ([PSS010962](#)) with an HR of 1.64 (95% CI: 1.09-2.48). Metric PPM018430 evaluated 8,982 South Asian ancestry individuals ([PSS010963](#)), yielding an HR of 1.6 (95% CI: 1.47-1.74).

Evaluated Samples

The UKB cohort provided four sample sets for evaluating coronary artery disease, all utilizing ICD 10 codes for phenotype definitions and reflecting a mean participant follow-up time of 12.0 years (range: 11.2 to 12.7 years). [PSS010960](#) evaluated 308,264 European individuals (45.6% male, mean age 57.3), explicitly excluding the first released tranche of genotypes from UKBB used in the training phase. The remaining evaluated sets included [PSS010961](#) for 7,281 African unspecified individuals (43.5% male, mean age 52.4), [PSS010962](#) for 1,464 East Asian individuals (37.2% male, mean age 53.0), and [PSS010963](#) for 8,982 South Asian individuals (54.1% male, mean age 53.8).

DATASET NAME: PGS003727

Polygenic Score (PGS) ID PGS003727 was released in the [PGS Catalog](#) on July 5, 2023. It is designed to predict coronary artery disease, which maps to the [coronary artery disorder](#) ontology term ([MONDO_0005010](#)). The score, named GPS_CADEUR, was developed using the LDpred2 method with the specific parameters set to $p=0.018$, h^2 scale=1, and Sparse LD=F. It was built on the original genome build hg19 and includes 1,125,113 variants, utilizing an effect weight type of beta. The score's catalog publication ID is [PGP000466](#), corresponding to the original citation by [Patel AP et al. Nat Med \(2023\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 504,636 individuals, categorized entirely (100%) as European ancestry. During the score development and training phase, 116,649 individuals were utilized, also consisting of 100% European ancestry. Across the 6 evaluated sample sets, the ancestry distribution is diverse: 33.3% European, 16.7% African, 16.7% East Asian, 16.7% Multi-ancestry (including European), and 16.7% South Asian.

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from [Europe PMC: 36474045](#). This dataset included 504,636 European individuals derived from the CARDIoGRAMplusC4D cohort. For the score's development and training, data was sourced from [Europe PMC: 30305743](#), utilizing 116,649 European (British) individuals (47.5% male) with a mean age of 57.5 years from the UKB cohort. Phenotypes in this training group were defined using various ICD 10 codes for myocardial infarction and related conditions (e.g., I21, I22, I23), specifically utilizing the first released tranche of genotypes from UKBB.

Performance Metrics

Patel AP et al. (2023) reported several performance metrics for predicting coronary artery disease, uniformly adjusting for age, sex, and the first ten principal components of genetic ancestry. Metric PPM018431 evaluated 308,264 European individuals ([PSS010960](#)), achieving a Hazard Ratio (HR) of 1.67. Metric PPM018432 evaluated 7,281 African ancestry individuals ([PSS010961](#)) with an HR of 1.15. Metric PPM018433 evaluated 1,464 East Asian ancestry individuals ([PSS010962](#)) with an HR of 1.54. Metric PPM018434 evaluated 8,982 South Asian ancestry individuals ([PSS010963](#)), yielding an HR of 1.57.

Additionally, Reeskamp LF et al. (2023) evaluated incident coronary artery disease among heterozygous familial hypercholesterolemia variant carriers, adjusting for age, sex, and 5 PCs. Metric PPM022290 assessed 1,315 European individuals ([PSS011902](#)) resulting in an HR of 1.35. Metric PPM022291 evaluated 429 Multi-ancestry individuals ([PSS011903](#)) yielding an HR of 1.31.

Evaluated Samples

The UKB cohort provided several sample sets for evaluating coronary artery disease, utilizing ICD 10 codes for phenotype definitions and reflecting a mean participant follow-up time of 12.0 years (range: 11.2 to 12.7 years). [PSS010960](#) evaluated 308,264 European individuals (45.6% male, mean age 57.3), explicitly excluding the first released tranche of genotypes from UKBB used in the training phase. The remaining UKB sets evaluated by Patel AP et al. included [PSS010961](#) for 7,281 African unspecified individuals (43.5% male, mean age 52.4), [PSS010962](#) for 1,464 East Asian individuals (37.2% male, mean age 53.0), and [PSS010963](#) for 8,982 South Asian individuals (54.1% male, mean age 53.8).

For the Reeskamp LF et al. evaluations, two additional datasets were used. The Dutch HeFH cohort ([PSS011902](#)) evaluated 1,315 European (Dutch) individuals (47.1% male) with a mean age of 42.1 years. Finally, [PSS011903](#) utilized an independent subset of the UKB cohort not used to train the PGS. This set comprised a total of 429 individuals, breaking down into 383 European, 27 Asian unspecified, 9 African American or Afro-Caribbean, and 10 Not reported ancestry individuals.

DATASET NAME: PGS004696

Polygenic Score (PGS) ID PGS004696 was released in the [PGS Catalog](#) on March 18, 2024. It predicts coronary heart disease, which maps to the [coronary artery disorder](#) ontology term ([MONDO_0005010](#)). The score, named multi_anc_hg37CSx, was developed using the PRS-CSx method with the specific parameter of global shrinkage $\phi=1e-2$. It is built on the original genome build hg19 and includes 1,289,980 variants, utilizing an effect weight type of beta. The score's catalog publication ID is [PGP000602](#), corresponding to the original citation by [Smith JL et al. Circ Genom Precis Med \(2024\)](#).

Ancestry Distribution

The variant associations (GWAS) were derived from a total of 1,093,078 individuals. The ancestry breakdown for this GWAS group is 70.7% European, 19.4% East Asian, 7% African, and 2.8% Hispanic or Latin American. During the score development and training phase, 87,724 individuals were utilized, consisting of 64.2% European, 19.4% African, 11.4% Hispanic or Latin American, and 5% Multi-ancestry (excluding European) individuals. Across the 5 evaluated sample sets, the ancestry distribution is perfectly balanced at 20% each for African, Hispanic or Latin American, East Asian, European, and South Asian ancestries.

Development Samples

The Genome-Wide Association Studies (GWAS) sourced data from GWAS Catalog: [GCST90132305](#) and Europe PMC: [35915156](#). The cohorts used included BBJ, CARDIoGRAMplusC4D, MVP, and UKB, which contributed 773,268 European individuals, 212,453 East Asian individuals, 76,709 African American or Afro-Caribbean individuals, and 30,648 Hispanic or Latin American individuals. For the score's development and training, data was drawn exclusively from the MVP cohort. This training group consisted of 56,359 European individuals, 16,987 African unspecified individuals, 9,992 Hispanic or Latin American individuals, and 4,386 East Asian and South Asian individuals (of which approximately 95% were of East Asian descent).

Performance Metrics

Smith JL et al. (2024) reported several performance metrics evaluating incident coronary heart disease, with all models uniformly adjusting for age, sex, and 10 principal components. Metric PPM020904 evaluated 52,702 European Ancestry individuals ([PSS011448](#)), achieving an Odds Ratio (OR) of 1.65 (95% CI: 1.59, 1.71) and an AUROC of 0.774. Metric PPM020906 evaluated 17,008 African Ancestry individuals ([PSS011446](#)) with an OR of 1.2 (95% CI: 1.15, 1.26) and an AUROC of 0.736. Metric PPM020908 evaluated 6,138 Hispanic or Latin American Ancestry individuals ([PSS011449](#)), reporting an OR of 1.51 (95% CI: 1.35, 1.69) and an AUROC of 0.706. Metric PPM020910 evaluated 22,751 East Asian Ancestry individuals ([PSS011447](#)) with an OR of 1.59 (95% CI: 1.54, 1.64) and an AUROC of 0.762. Finally, metric PPM020912 evaluated 9,178 South Asian Ancestry individuals ([PSS011450](#)), yielding an OR of 2.67 (95% CI: 2.39, 3.01) and an AUROC of 0.803.

Evaluated Samples

Five distinct sample sets were analyzed for the evaluation phase. [PSS011446](#) evaluated 17,008 African unspecified individuals using data from the ARIC, CHS, MESA, WHI, and eMERGE cohorts. [PSS011447](#) evaluated 22,751 East Asian individuals using data from the BBJ, OACIS, and TaiChi cohorts. [PSS011448](#) assessed 52,702 European individuals sourced from the ARIC, CHS, MESA, WHI, and eMERGE cohorts. [PSS011449](#) evaluated 6,138 Hispanic or Latin American individuals drawn from the MESA, WHI, and eMERGE cohorts. Lastly, [PSS011450](#) evaluated 9,178 South Asian individuals sourced exclusively from the UKB cohort.

DATASET NAME: PGS004698

Polygenic Score (PGS) ID PGS004698, which was released in the [PGS Catalog](#) on March 18, 2024. The primary predicted trait is reported as Coronary heart disease, which maps to the broader ontological trait of coronary artery disorder (MONDO_0005010). The score is explicitly designated as providing multi-ancestry trait information.

Score Construction and Source

The polygenic score is named "multi_anc_hg37PT". It was developed utilizing the Pruning and Thresholding (P+T) method. The specific statistical parameters applied during development included a P-value threshold of less than or equal to 0.1, an LD pruning window distance of 250 kb, and an R-squared value of less than or equal to 0.8. The score variants were mapped using the original genome build hg19 and comprise a total of 542,218 variants. The effect weight type calculated for these variants is beta. The primary source publication for this score is documented under PGP ID PGP000602, citing the study by Smith JL et al., published in Circ Genom Precis Med in 2024.

Ancestry Distribution Overview

The genetic data was sourced from three distinct phases: variant associations (GWAS), score development/training, and evaluation, each with varying ancestry distributions.

- **GWAS Phase:** The source of variant associations utilized a massive pool of 1,093,078 individuals. The ancestry breakdown was predominantly European at 70.7%, followed by East Asian at 19.4%, African at 7%, and Hispanic or Latin American at 2.8%.
- **Training Phase:** For score development and training, 87,724 individuals were used. The distribution was 64.2% European, 19.4% African, 11.4% Hispanic or Latin American, and 5% Multi-ancestry (excluding European).
- **Evaluation Phase:** The evaluation of the PGS was conducted evenly across 5 sample sets, with African, Hispanic or Latin American, East Asian, European, and South Asian ancestries each representing 20% of the evaluation sample pools.

Development Samples (GWAS) Details

The source of variant associations (GWAS) corresponds to the study identifiers GCST90132305 (GWAS Catalog) and 35915156 (Europe PMC). The data was aggregated from four major cohorts: BBJ, CARDIoGRAMplusC4D, MVP, and UKB. Within these cohorts, the sampling included 773,268 European individuals, 30,648 Hispanic or Latin American individuals, 76,709 African American or Afro-Caribbean individuals, and 212,453 East Asian individuals.

Score Development and Training Details

During the score development and training phase, data was drawn exclusively from the MVP cohort. The training subsets consisted of 56,359 European individuals, 16,987 individuals of unspecified African ancestry, and 9,992 Hispanic or Latin American individuals. Additionally, a combined group of 4,386 individuals of East Asian and South Asian descent was utilized, with approximately 95% of this specific subset being of East Asian ancestry.

Performance Metrics and Evaluated Samples

The performance of the PGS was evaluated on the reported trait of incident coronary heart disease across five distinct ancestral sample sets. In all evaluations, the statistical models adjusted for covariates including age, sex, and 10 principal components (PCs).

European Evaluation (Metric ID: PPM020913): This evaluation used Sample Set PSS011448, consisting of 52,702 individuals of European ancestry from the ARIC, CHS, MESA, WHI, and eMERGE cohorts. The score demonstrated an Odds Ratio (OR) per standard deviation change of 1.65 (confidence interval: 1.59-1.72) and an AUROC classification metric of 0.773.

African Evaluation (Metric ID: PPM020914): This evaluation used Sample Set PSS011446, comprising 17,008 individuals of unspecified African ancestry sourced from the ARIC, CHS, MESA, WHI, and eMERGE cohorts. The resulting performance showed an OR of 1.16 (confidence interval: 1.11-1.21) and an AUROC of 0.735.

Hispanic or Latin American Evaluation (Metric ID: PPM020915): Using Sample Set PSS011449, 6,138 individuals of Hispanic or Latin American ancestry from the MESA, WHI, and eMERGE cohorts were evaluated. The PGS yielded an OR of 1.38 (confidence interval: 1.24-1.54) and an AUROC of 0.699.

East Asian Evaluation (Metric ID: PPM020916): This analysis utilized Sample Set PSS011447, testing 22,751 East Asian individuals from the BBJ, OACIS, and TaiChi cohorts. The performance metrics indicated an OR of 1.56 (confidence interval: 1.5-1.61) alongside an AUROC of 0.748.

South Asian Evaluation (Metric ID: PPM020917): Finally, Sample Set PSS011450 evaluated 9,178 individuals of South Asian ancestry entirely from the UKB cohort. This sample set showed a substantial OR of 2.75 (confidence interval: 2.41-3.14) and the highest AUROC among the sets at 0.786.

DATASET NAME: PGS004879

Polygenic Score (PGS) ID PGS004879, which was released in the PGS Catalog on June 27, 2024. The primary predicted trait is reported as Coronary heart disease, which maps to the broader ontological trait of [coronary artery disorder \(MONDO_0005010\)](#).

Score Construction and Source

The polygenic score is named "INTERVENE_MegaPRS_CHD". It was developed utilizing the "megaprs.auto" method. The specific statistical parameters applied during development used the BayesR model with parameters set to $p1=0.89$, $p2=0.05$, $p3=0.05$, and $p4=0.01$. The score variants were mapped using the original genome build GRCh37 and comprise a total of 610,677 variants. The effect weight type calculated for these variants is beta. The primary source publication for this score is documented under [PGP000618](#), citing the study by Jermy B et al., published in Nat Commun (2024).

Ancestry Distribution Overview

The genetic data was sourced from three distinct phases: variant associations (GWAS), score development/training, and evaluation.

- **GWAS Phase:** The source of variant associations utilized a multi-ancestry (including European) distribution, accounting for 100% of the 63,731 individuals in this phase.
- **Training Phase:** For score development and training, 404 individuals were used, consisting of 100% European ancestry.
- **Evaluation Phase:** The evaluation of the PGS was conducted entirely on individuals of European ancestry (100%) across 7 distinct sample sets.

Development Samples (GWAS) Details

The source of variant associations (GWAS) corresponds to the study identifiers [GCST004787](#) (GWAS Catalog) and [28714975](#) (Europe PMC). The data aggregates a sample size of 63,731 individuals identified broadly as having European ancestry.

Score Development and Training Details

During the score development and training phase, data was drawn exclusively from the 1000G cohort. The training subset consisted of 404 individuals specifically noted as being non-Finnish Europeans.

Performance Metrics and Evaluated Samples

The performance of the PGS was evaluated on the reported trait of incident coronary heart disease (CHD) across seven distinct sample sets of European descent. In all evaluations, the statistical models adjusted for covariates using principal components (PCs) 1-10.

- **MGBB Cohort Evaluation (Metric ID: PPM021209):** This evaluation used Sample Set [PSS011573](#), consisting of 38,448 individuals from the MGBB cohort. The score demonstrated a Hazard Ratio (HR) per standard deviation change of 1.18 (confidence interval: 1.16-1.2) and a C-index classification metric of 0.55 (confidence interval: 0.55-0.56).
- **HUNT Cohort Evaluation (Metric ID: PPM021210):** This evaluation used Sample Set [PSS011572](#), comprising 69,715 individuals from the HUNT cohort. The resulting performance showed an HR of 1.39 (confidence interval: 1.36-1.42) and a C-index of 0.6 (confidence interval: 0.59-0.6).
- **GEL Cohort Evaluation (Metric ID: PPM021211):** Using Sample Set [PSS011571](#), 29,427 individuals from the GEL cohort were evaluated. The PGS yielded an HR of 1.32 (confidence interval: 1.27-1.38) and a C-index of 0.6 (confidence interval: 0.58-0.61).
- **G&H Cohort Evaluation (Metric ID: PPM021212):** This analysis utilized Sample Set [PSS011569](#), testing 44,168 individuals from the G&H cohort. The performance metrics indicated an HR of 1.27 (confidence interval: 1.23-1.32). No C-index was reported for this sample set.
- **GS:SFHS Cohort Evaluation (Metric ID: PPM021213):** Sample Set [PSS011570](#) evaluated 7,018 individuals from the GS:SFHS cohort. This set showed an HR of 1.13 (confidence interval: 1.07-1.19) and a C-index of 0.56 (confidence interval: 0.54-0.57).
- **FinnGen Cohort Evaluation (Metric ID: PPM021214):** Tested on a large subset, Sample Set [PSS011568](#) evaluated 412,090 individuals from the FinnGen cohort. It demonstrated the highest classification metrics among the sets, with an HR of 1.41 (confidence interval: 1.4-1.43) and a C-index of 0.62 (confidence interval: 0.62-0.62).
- **EB Cohort Evaluation (Metric ID: PPM021215):** Finally, Sample Set [PSS011567](#) included 148,312 individuals from the EB cohort. The evaluation yielded an HR of 1.18 (confidence interval: 1.16-1.21) alongside a C-index of 0.58 (confidence interval: 0.57-0.58).

DATASET NAME: PGS003725

Polygenic Score (PGS) with the ID **PGS004888**. This score was officially released in the PGS Catalog on May 8, 2024. The reported predicted trait for this score is "Coronary artery disease" [sic], which maps to the formal trait of "coronary artery disorder" under the ontology identifier MONDO_0005010.

Score Details and Construction

The Polygenic Score is formally named **CAD_gePGS**. It was constructed using the **PRS-CS** development method. The specific parameters for its development utilized PRS-CS-auto, leveraging a 1000G European sample of 503 individuals as the external linkage disequilibrium (LD) reference panel.

Regarding its genetic variants, the score was built on the original genome build **GRCh37** and consists of exactly 1,110,046 variants. The effect weight type utilized for the score is categorized as "beta".

The source of this Polygenic Score in the PGS Catalog carries the Publication (PGP) ID **PGP000619** and is cited from the publication by Mandla R et al. in *Genome Med* (2024).

Ancestry Distribution Summary

The distribution of ancestries changes across the three primary phases of the score's life cycle:

- **Source of Variant Associations (GWAS):** This phase utilized a 100% multi-ancestry group (including European populations), totaling 63,731 individuals.
- **Score Development/Training:** This phase utilized 53,555 individuals. The ancestry breakdown is 84% European, 6% Hispanic or Latin American, 4.9% African, 1.7% East Asian, 1.5% Not Reported, 1.1% South Asian, and 0.9% Greater Middle Eastern.
- **PGS Evaluation:** The evaluation was conducted on a single sample set composed entirely (100%) of a multi-ancestry group (including European populations).

Development Samples

Source of Variant Associations (GWAS) The Genome-Wide Association Study (GWAS) data was sourced from study identifiers GCST004787 (GWAS Catalog) and 28714975 (Europe PMC). This dataset included 63,731 individuals of European and Not Reported (NR) ancestries. Specific cohort information was not reported for this dataset.

Score Development and Training Breakdown For the score development and training phase, all participants were drawn exclusively from the MGBB cohort. The sample pool consisted of seven distinct subgroups based on ancestry:

- 44,964 individuals of European ancestry.
- 3,199 individuals of Hispanic or Latin American ancestry.
- 2,628 individuals of unspecified African ancestry.
- 902 individuals of East Asian ancestry.
- 825 individuals whose ancestry was not reported.
- 573 individuals of South Asian ancestry.
- 464 individuals of Greater Middle Eastern ancestry (which includes Middle Eastern, North African, or Persian).

Performance Metrics

Note: The PGS Catalog states a disclaimer that performance metrics are displayed exactly as reported by the source studies. These metrics are not necessarily comparable with each other, as they depend heavily on sample characteristics, phenotyping, and statistical modeling.

The performance of the score was measured in two different clinical scenarios, both sourced from the Mandla R et al. *Genome Med* (2024) publication (PGP000619). Both metrics were evaluated on the same sample set (PSS011669), which consists of 11,515 individuals from a multi-ancestry background (including European). In both evaluations, the statistical model included the following covariates: the first 10 ancestry principal components (PCs), age, sex, smoking status, systolic blood pressure, HDL, and total cholesterol.

- **First Metric (PPM021266):** This metric evaluated the reported trait of incident coronary artery disease cases *without diabetes*. It yielded a PGS Effect Size (Hazard Ratio per standard deviation change) of 1.27, with a confidence interval of [1.18, 1.36]. The classification metric measured a C-index of 0.752.
- **Second Metric (PPM021268):** This metric evaluated the reported trait of incident coronary artery disease cases *with diabetes*. It yielded a PGS Effect Size (Hazard Ratio per standard deviation change) of 1.13, with a confidence interval of [1.02, 1.25]. The classification metric measured a C-index of 0.668.

Evaluated Samples

The evaluation phase relied on the PGS Sample Set ID **PSS011669**, with all individuals sourced from the MGBB cohort. Across all ancestry subgroups in this sample set, the participant profile remained consistent: a mean age of 60.4 years, a median follow-up time of 13.9 years, and exactly 42.32% of the samples being male. The phenotype definition and inclusion criteria for all groups required an MGH encounter, an ICD-9 code for diabetes, and a problem list term for diabetes, or specific HbA1C criteria.

The 11,515 individuals in this evaluated sample set are divided by ancestry as follows:

- 9,909 individuals of European ancestry.
- 757 individuals of Hispanic or Latin American ancestry.
- 347 individuals of unspecified African ancestry.
- 183 individuals of East Asian ancestry.
- 154 individuals whose ancestry was not reported.
- 83 individuals of Greater Middle Eastern ancestry (Middle Eastern, North African, or Persian).
- 82 individuals of South Asian ancestry.

DATASET NAME: PGS005091

The score was officially published in the PGS Catalog on November 21, 2024. The reported predicted trait for this score is **Coronary artery disease**, which maps formally to **coronary artery disorder** in the ontology under identifier MONDO_0005010.

Score Details and Construction

The formal name of this Polygenic Score is **PGS_LDP2Auto**. It was developed using the **LDPred2Auto** methodology; specific parameter settings for the run were Not Reported (NR).

The score was constructed on the **GRCh38** original genome build and incorporates a total of **1,428,772 genetic variants**. The effect weight type utilized across these variants is categorized as **beta**.

The source publication for this score in the PGS Catalog carries the Publication ID **PGP000684**, citing the research paper by Abramowitz SA et al., published in *JAMA* (2024).

Ancestry Distribution Summary

The breakdown of participant ancestries across the development and evaluation phases is structured as follows:

- **Source of Variant Associations (GWAS):** Sourced from a total population of **2,216,963 individuals** (100%). The ancestral composition consists of 52.6% Multi-ancestry (including European), 29.5% European, 13.1% East Asian, 3.5% African, and 1.4% Hispanic or Latin American.
 - **PGS Evaluation:** Conducted across **3 distinct sample sets**, where 100% of the evaluated populations were multi-ancestry groups (including European populations).
-

Development Samples: Source of Variant Associations (GWAS)

The Genome-Wide Association Study (GWAS) dataset encompasses 2,216,963 individuals partitioned across seven study groups:

- **GCST90132314 / Europe PMC 36474045:** Comprises 1,165,690 individuals of European and Not Reported (NR) ancestries. Cohort and case/control breakdowns were Not Reported.
- **FinnGen Cohort (PMID: 36653562):** Consists of 360,950 individuals of European ancestry, partitioned into 47,550 cases and 313,400 controls.
- **Million Veteran Program / MVP Cohort (PMID: 35915156):** Split into three ancestry cohorts:
 - 292,438 individuals of European ancestry (95,151 cases, 197,287 controls).
 - 76,709 individuals of unspecified African ancestry (17,202 cases, 59,507 controls).
 - 30,648 individuals of Hispanic or Latin American ancestry (6,378 cases, 24,270 controls).
- **BioBank Japan / BBJ Cohort (PMID: 32514122):** Consists of 212,453 individuals of East Asian ancestry, broken down into 29,319 cases and 183,134 controls.

- **China Kadoorie Biobank / CKB Cohort (PMID: 37601966):** Consists of 78,075 individuals of East Asian ancestry, comprising 3,386 cases and 74,689 controls.
-

Performance Metrics

Note: The PGS Catalog provides a disclaimer that performance metrics are presented exactly as reported in the primary study (Abramowitz SA et al., JAMA 2024 / PGP000684) and are not directly comparable across different sample sets due to variations in phenotyping, cohort characteristics, and covariate modeling. In all three performance evaluations below, the predicted trait evaluated was "Prevalent coronary heart disease," and the polygenic scores were PCA-normalized in the models.

- **Metric PPM022178 (Evaluated on Sample Set PSS011830):** Evaluated on a multi-ancestry sample of 171,095 individuals with age and sex included as covariates. The model produced a PGS Effect Size (Odds Ratio per SD change) of 1.452 with a 95% confidence interval of [1.427, 1.477], a classification AUROC of 0.776, and a Brier score of 0.0825.
 - **Metric PPM022179 (Evaluated on Sample Set PSS011832):** Evaluated on a multi-ancestry sample of 53,092 individuals with age, sex, and the first 5 principal components (PCs) included as covariates. The model produced an Odds Ratio of 1.374 [1.343, 1.406], an AUROC of 0.8, and a Brier score of 0.143.
 - **Metric PPM022180 (Evaluated on Sample Set PSS011831):** Evaluated on a multi-ancestry sample of 41,193 individuals with age and sex included as covariates. The model produced an Odds Ratio of 1.547 [1.506, 1.59], an AUROC of 0.791, and a Brier score of 0.1405.
-

Evaluated Samples (Complete 18-Subgroup Breakdown)

The evaluation phase spans 3 primary sample sets encompassing a total of 18 ancestry subgroups. For all sample sets, participant follow-up time, age distributions, and additional cohort information were not specified.

1. Sample Set PSS011830 (AllofUs Cohort)

- **Total Sample Size:** 171,095 individuals.
- **Phenotype Definition:** Based on ICD-9 codes (410, 411, 412, 413, 41, V45.81) and ICD-10 codes (I21, I22, I24, Z95.1, Z98.61, I20.0).
- **Ancestry Subgroup Breakdown:**
 - **European:** 100,493 individuals (11,344 cases, 89,149 controls).
 - **African Unspecified:** 35,590 individuals (3,558 cases, 32,032 controls).

- **Hispanic or Latin American:** 29,801 individuals (2,268 cases, 27,533 controls).
- **East Asian:** 1,997 individuals (132 cases, 1,865 controls).
- **South Asian:** 1,746 individuals (134 cases, 1,612 controls).
- **Greater Middle Eastern (Middle Eastern, North African, or Persian):** 1,468 individuals (153 cases, 1,315 controls).

2. Sample Set PSS011831 (PMB Cohort)

- **Total Sample Size:** 41,193 individuals.
- **Phenotype Definition:** Identical to PSS011830, utilizing ICD-9 codes (410, 411, 412, 413, 41, V45.81) and ICD-10 codes (I21, I22, I24, Z95.1, Z98.61, I20.0).
- **Ancestry Subgroup Breakdown:**
 - **European:** 28,084 individuals (6,761 cases, 21,323 controls).
 - **African Unspecified:** 9,830 individuals (1,932 cases, 7,898 controls).
 - **Hispanic or Latin American:** 1,877 individuals (312 cases, 1,565 controls).
 - **East Asian:** 584 individuals (70 cases, 514 controls).
 - **South Asian:** 550 individuals (88 cases, 462 controls).
 - **Greater Middle Eastern (Middle Eastern, North African, or Persian):** 268 individuals (52 cases, 216 controls).

3. Sample Set PSS011832 (UCLA Cohort)

- **Total Sample Size:** 53,092 individuals.
- **Phenotype Definition:** Defined using Chronic Condition Warehouse definitions for myocardial infarction and ischemic heart disease.
- **Ancestry Subgroup Breakdown:**
 - **European:** 33,714 individuals (8,137 cases, 25,577 controls).
 - **Hispanic or Latin American:** 7,516 individuals (1,510 cases, 6,006 controls).
 - **East Asian:** 4,959 individuals.
 - **Not Reported:** 3,413 individuals.
 - **African Unspecified:** 2,682 individuals (731 cases, 1,951 controls).
 - **South Asian:** 808 individuals.

DATASET NAME: PGS005092

PGS005092. The score was officially released in the PGS Catalog on November 21, 2024. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term [MONDO_0005010](#).

Score Details and Construction

The formal name of this Polygenic Score is **PGS_prscsx**. It was constructed utilizing the **PRS-CSx** development method, although specific parameters for this development process are listed as Not Reported (NR).

The score was built upon the original genome build **GRCh38** and is composed of exactly **1,279,502 variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID [PGP000684](#). This score originates from the publication cited as [Abramowitz SA et al. JAMA \(2024\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and testing is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an extensive pool of 2,216,963 individuals (accounting for 100% of the GWAS pool). The overall breakdown comprises 52.6% from a multi-ancestry group (including European populations), 29.5% European, 13.1% East Asian, 3.5% African, and 1.4% Hispanic or Latin American.
 - **PGS Evaluation:** The evaluation phase was conducted across 3 distinct sample sets. The participant pool in this phase consisted entirely (100%) of individuals from multi-ancestry groups (including European populations).
-

Development Samples: Source of Variant Associations (GWAS)

The Genome-Wide Association Study (GWAS) data was sourced from multiple studies and cohorts, yielding a combined 2,216,963 individuals across seven subsets:

- A combined dataset documented under GWAS Catalog [GCST90132314](#) and Europe PMC [36474045](#) provided 1,165,690 individuals of European and Not Reported (NR) ancestries. The exact case/control numbers and cohort name were Not Reported (NR).
- The **FinnGen** cohort (PMID: 36653562) contributed 360,950 individuals of European ancestry. This group is partitioned into 47,550 cases and 313,400 controls.
- The **MVP** cohort (PMID: 35915156) provided data across three distinct ancestry groups: 292,438 individuals of European ancestry (95,151 cases, 197,287 controls); 76,709 individuals of unspecified African ancestry (17,202 cases, 59,507 controls); and 30,648 individuals of Hispanic or Latin American ancestry (6,378 cases, 24,270 controls).

- The **BBJ** cohort (PMID: 32514122) contributed 212,453 individuals of East Asian ancestry, comprising 29,319 cases and 183,134 controls.
 - The **CKB** cohort (PMID: 37601966) contributed 78,075 individuals of East Asian ancestry, comprising 3,386 cases and 74,689 controls.
-

Performance Metrics

Disclaimer from the PGS Catalog: Performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated for the reported trait of "Prevalent coronary heart disease" across three separate multi-ancestry sample sets sourced from [PGP000684 \(Abramowitz SA et al. JAMA \(2024\)\)](#). In all three evaluated models, the scores were PCA-normalized.

- **Metric PPM022181:** Evaluated on the [PSS011830](#) sample set (171,095 individuals) with age and sex included as covariates. The evaluation yielded a PGS Effect Size (Odds Ratio per standard deviation change) of 1.407, with a 95% confidence interval of [1.384, 1.43]. The classification metric measured an AUROC of 0.774, alongside a Brier score of 0.0828.
 - **Metric PPM022182:** Evaluated on the [PSS011832](#) sample set (53,092 individuals) with age, sex, and the first 5 principal components (PCs) included as covariates. The evaluation yielded an Odds Ratio of 1.384, with a confidence interval of [1.353, 1.416]. The classification metric measured an AUROC of 0.801, alongside a Brier score of 0.1425.
 - **Metric PPM022183:** Evaluated on the [PSS011831](#) sample set (41,193 individuals) with age and sex included as covariates. The evaluation yielded an Odds Ratio of 1.562, with a confidence interval of [1.522, 1.604]. The classification metric measured an AUROC of 0.793, alongside a Brier score of 0.1402.
-

Evaluated Samples

The evaluation phase relied on three distinct multi-ancestry sample sets, breaking down into a total of 18 ancestry subgroups. For all sample sets and their respective subgroups, participant follow-up time, participant age, and additional sample/cohort information were not reported.

1. Sample Set [PSS011830](#) (AllofUs Cohort)

- **Total Sample Size:** 171,095 individuals.
- **Phenotype Definition:** Based on ICD-9 codes (410, 411, 412, 413, 41, V45.81) and ICD-10 codes (I21, I22, I24, Z95.1, Z98.61, I20.0).

- **Ancestry Subgroup Breakdown:**
 - **European:** 100,493 individuals (11,344 cases, 89,149 controls)
 - **African Unspecified:** 35,590 individuals (3,558 cases, 32,032 controls)
 - **Hispanic or Latin American:** 29,801 individuals (2,268 cases, 27,533 controls)
 - **East Asian:** 1,997 individuals (132 cases, 1,865 controls)
 - **South Asian:** 1,746 individuals (134 cases, 1,612 controls)
 - **Greater Middle Eastern (Middle Eastern, North African, or Persian):** 1,468 individuals (153 cases, 1,315 controls)

2. Sample Set [PSS011831](#) (PMB Cohort)

- **Total Sample Size:** 41,193 individuals.
- **Phenotype Definition:** Identical to PSS011830, based on ICD-9 codes (410, 411, 412, 413, 41, V45.81) and ICD-10 codes (I21, I22, I24, Z95.1, Z98.61, I20.0).
- **Ancestry Subgroup Breakdown:**
 - **European:** 28,084 individuals (6,761 cases, 21,323 controls)
 - **African Unspecified:** 9,830 individuals (1,932 cases, 7,898 controls)
 - **Hispanic or Latin American:** 1,877 individuals (312 cases, 1,565 controls)
 - **East Asian:** 584 individuals (70 cases, 514 controls)
 - **South Asian:** 550 individuals (88 cases, 462 controls)
 - **Greater Middle Eastern (Middle Eastern, North African, or Persian):** 268 individuals (52 cases, 216 controls)

3. Sample Set [PSS011832](#) (UCLA Cohort)

- **Total Sample Size:** 53,092 individuals.
- **Phenotype Definition:** Defined using Chronic Condition Warehouse definitions for myocardial infarction and ischemic heart disease.
- **Ancestry Subgroup Breakdown:**
 - **European:** 33,714 individuals (8,137 cases, 25,577 controls)
 - **Hispanic or Latin American:** 7,516 individuals (1,510 cases, 6,006 controls)
 - **East Asian:** 4,959 individuals
 - **Not Reported:** 3,413 individuals
 - **African Unspecified:** 2,682 individuals (731 cases, 1,951 controls)
 - **South Asian:** 808 individuals

DATASET NAME: PGS005112

PGS005112. The score was officially released in the PGS Catalog on **February 4, 2025**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term [MONDO_0005010](#).

Score Details and Construction

The formal name of this Polygenic Score is **PGS_CAD**. It was constructed utilizing the **PRS-CSx** development method, with the specific parameter configured as **PHI=AUTO**.

The score was built upon the original genome build **GRCh37** and consists of exactly **1,106,628 genetic variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID [PGP000701](#). This score originates from the publication cited as [Loesch DP et al. Nat Commun \(2025\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **400,052 individuals**, representing 100% of the dataset. The ancestral composition consists of 55.9% East Asian, 35.3% European, 6.4% South Asian, 1% Hispanic or Latin American, 0.8% African, and 0.6% Greater Middle Eastern populations.
 - **PGS Evaluation:** The evaluation phase was conducted across **2 distinct sample sets**. The participant breakdown for this phase consists of an even split: 50% European ancestry and 50% Multi-ancestry (which includes European populations).
-

Development Samples: Source of Variant Associations (GWAS)

The Genome-Wide Association Study (GWAS) dataset encompasses 400,052 individuals sourced from two primary cohorts across seven specific study groups:

1. CARDIoGRAMplusC4D Cohort (Study IDs: [GCST003116](#) / Europe PMC: [26343387](#))

This cohort contributed data across six distinct ancestry groups:

- 141,217 individuals of **European** ancestry.
- 25,557 individuals of **South Asian** ancestry.
- 11,323 individuals of **East Asian** ancestry.
- 4,095 individuals of **Hispanic or Latin American** ancestry.
- 3,139 individuals of **African American or Afro-Caribbean** ancestry.
- 2,268 individuals of **Greater Middle Eastern** (Middle Eastern, North African, or Persian) ancestry.

2. BBJ Cohort (Study IDs: [GCST90013687](#) / Europe PMC: [32514122](#)) This cohort contributed a single subset to the GWAS data:

- 212,453 individuals of **East Asian** ancestry.
-

Performance Metrics

Disclaimer from the PGS Catalog: Performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated for the reported trait of "coronary artery disease" across two separate sample sets, both sourced from [PGP000701](#) (Loesch DP *et al.* Nat Commun (2025)). In both evaluated models, the statistical covariates included: age, age², sex, UKB centre, array, and genetic PCs 1-20. Classification metrics and other metrics (like AUROC or Brier scores) were not reported for these models.

- **Metric PPM022303:** Evaluated on the [PSS011911](#) sample set (Multi-ancestry including European, totaling 48,603 individuals). The evaluation yielded a PGS Effect Size (Odds Ratio per standard deviation change) of **1.45**, with a 95% confidence interval of **[1.41, 1.49]**.
 - **Metric PPM022314:** Evaluated on the [PSS011912](#) sample set (European Ancestry, totaling 45,356 individuals). The evaluation yielded a PGS Effect Size (Odds Ratio) of **1.46**, with a confidence interval of **[1.42, 1.5]**.
-

Evaluated Samples

The evaluation phase relied on two distinct sample sets drawn from the **UKB (UK Biobank)** cohort. Both sample sets share identical participant follow-up times of **Median = 14.8 years** and a sex distribution of **46.0% Male samples**. No additional ancestry description or other sample/cohort information was reported.

1. Sample Set [PSS011911](#)

- **Total Sample Size:** 48,603 individuals.
- **Phenotype Definitions and Methods:** Defined as ICD10: E11 for T2D, N18 for CKD, first occurrence of I20-I25 for CAD. BMI is defined as baseline body mass index.
- **Participant Age:** Mean = 56.8 years, Standard Deviation (Sd) = 8.2 years.
- **Sample Ancestry:** Multi-ancestry, specifically including European, East Asian, South Asian, Hispanic or Latin American, and African unspecified.
- **Cohort:** UKB

2. Sample Set [PSS011912](#)

- **Total Sample Size:** 45,356 individuals.
- **Phenotype Definitions and Methods:** Defined as ICD10: E11 for T2D, N18 for CKD, first occurrence of I20-I25 for CAD, K76.0 for NAFLD. BMI is defined as baseline body mass index.
- **Participant Age:** Mean = 57.1 years, Standard Deviation (Sd) = 8.1 years.
- **Sample Ancestry:** Exclusively European.
- **Cohort:** UKB

DATASET NAME: PGS005112

PGS005112. The score was officially released in the PGS Catalog on **February 4, 2025**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term [MONDO_0005010](#).

Score Details and Construction

The formal name of this Polygenic Score is **PGS_CAD**. It was constructed utilizing the **PRS-CSx** development method, with the specific parameter configured as **PHI=AUTO**.

The score was built upon the original genome build **GRCh37** and consists of exactly **1,106,628 genetic variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID [PGP000701](#). This score originates from the publication cited as [Loesch DP et al. Nat Commun \(2025\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **400,052 individuals**, representing 100% of the dataset. The ancestral composition consists of 55.9% East Asian, 35.3% European, 6.4% South Asian, 1% Hispanic or Latin American, 0.8% African, and 0.6% Greater Middle Eastern populations.

- **PGS Evaluation:** The evaluation phase was conducted across **2 distinct sample sets**. The participant breakdown for this phase consists of an even split: 50% European ancestry and 50% Multi-ancestry (which includes European populations).
-

Development Samples: Source of Variant Associations (GWAS)

The Genome-Wide Association Study (GWAS) dataset encompasses 400,052 individuals sourced from two primary cohorts across seven specific study groups:

1. CARDIoGRAMplusC4D Cohort (Study IDs: [GCST003116](#) / Europe PMC: [26343387](#))

This cohort contributed data across six distinct ancestry groups:

- 141,217 individuals of **European** ancestry.
- 25,557 individuals of **South Asian** ancestry.
- 11,323 individuals of **East Asian** ancestry.
- 4,095 individuals of **Hispanic or Latin American** ancestry.
- 3,139 individuals of **African American or Afro-Caribbean** ancestry.
- 2,268 individuals of **Greater Middle Eastern** (Middle Eastern, North African, or Persian) ancestry.

2. BBJ Cohort (Study IDs: [GCST90013687](#) / Europe PMC: [32514122](#))

This cohort contributed a single subset to the GWAS data:

- 212,453 individuals of **East Asian** ancestry.
-

Performance Metrics

Disclaimer from the PGS Catalog: Performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated for the reported trait of "coronary artery disease" across two separate sample sets, both sourced from [PGP000701](#) (Loesch DP *et al.* Nat Commun (2025)). In both evaluated models, the statistical covariates included: age, age², sex, UKB centre, array, and genetic PCs 1-20. Classification metrics and other metrics (like AUROC or Brier scores) were not reported for these models.

- **Metric PPM022303:** Evaluated on the [PSS011911](#) sample set (Multi-ancestry including European, totaling 48,603 individuals). The evaluation yielded a PGS Effect Size (Odds Ratio per standard deviation change) of **1.45**, with a 95% confidence interval of [**1.41, 1.49**].

- **Metric PPM022314:** Evaluated on the [PSS011912](#) sample set (European Ancestry, totaling 45,356 individuals). The evaluation yielded a PGS Effect Size (Odds Ratio) of **1.46**, with a confidence interval of **[1.42, 1.5]**.
-

Evaluated Samples

The evaluation phase relied on two distinct sample sets drawn from the **UKB (UK Biobank)** cohort. Both sample sets share identical participant follow-up times of **Median = 14.8 years** and a sex distribution of **46.0% Male samples**. No additional ancestry description or other sample/cohort information was reported.

1. Sample Set [PSS011911](#)

- **Total Sample Size:** 48,603 individuals.
- **Phenotype Definitions and Methods:** Defined as ICD10: E11 for T2D, N18 for CKD, first occurrence of I20-I25 for CAD. BMI is defined as baseline body mass index.
- **Participant Age:** Mean = 56.8 years, Standard Deviation (Sd) = 8.2 years.
- **Sample Ancestry:** Multi-ancestry, specifically including European, East Asian, South Asian, Hispanic or Latin American, and African unspecified.
- **Cohort:** UKB

2. Sample Set [PSS011912](#)

- **Total Sample Size:** 45,356 individuals.
- **Phenotype Definitions and Methods:** Defined as ICD10: E11 for T2D, N18 for CKD, first occurrence of I20-I25 for CAD, K76.0 for NAFLD. BMI is defined as baseline body mass index.
- **Participant Age:** Mean = 57.1 years, Standard Deviation (Sd) = 8.1 years.
- **Sample Ancestry:** Exclusively European.
- **Cohort:** UKB

DATASET NAME : PGS005296

PGS005296. The score was officially released in the PGS Catalog on **December 18, 2025**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term **MONDO_0005010**.

Score Details and Construction

The formal name of this Polygenic Score is **Jorda_CAD_SingleTrait-PGS_2024**. It was constructed utilizing the **LDpred2** development method. The specific parameters configured for this development method were a Grid method with ρ (rho) set to 0.0056, h^2 SNP set to 0.0207, and a sparsity of 0.

The score was built upon the original genome build **GRCh38** and consists of exactly **1,042,395 genetic variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID **PGP000759**. This score originates from the publication cited as [Jordà P et al. NPJ Genom Med \(2025\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **63,731 individuals**, representing 100% of the dataset. The ancestral composition for this phase was classified entirely (100%) as a Multi-ancestry group (including European populations).
- **Score Development/Training:** This phase utilized a pool of **2,500 individuals**, drawn entirely (100%) from populations of European ancestry.
- **PGS Evaluation:** The evaluation phase was conducted across **1 Sample Set**, with 100% of the participants classified as having European ancestry.

Development Samples

The development phase relied on two distinct sample sets—one for the initial GWAS and one for score development/training.

1. Source of Variant Associations (GWAS) The Genome-Wide Association Study (GWAS) dataset encompasses **63,731 individuals**. The sample ancestry is listed as European and Not Reported (NR). The specific cohort was also Not Reported (NR). This data was sourced from study identifiers in the GWAS Catalog ([GCST004787](#)) and Europe PMC ([28714975](#)).

2. Score Development/Training The training dataset encompasses **2,500 individuals** of exclusively European ancestry, sourced from the **MHI** cohort. The study identifier for this dataset is found in Europe PMC under the DOI [10.1038/s41525-025-00515-2](#). The phenotype definitions and methods for this group were based on "Atrial fibrillation or flutter." Details regarding the age of study participants, participant follow-up time, additional ancestry descriptions, and additional sample/cohort information were not reported.

Performance Metrics

Note: The PGS Catalog includes a disclaimer that performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated under a single metric ID, **PPM023004**, sourced from the Jordà P *et al.* NPJ Genom Med (2025) publication ([PGP000759](#)).

The model was evaluated for the reported trait of "Prevalent coronary artery disease defined as myocardial infarction, percutaneous transluminal coronary angioplasty, coronary artery bypass grafting, unstable angina and chronic ischemic heart disease." The statistical covariates included in the model were age, sex, and the four first principal components of genetic ancestry.

The evaluation yielded a classification metric **AUROC of 0.772** and an **R² (Other Metric) of 0.27809**. The PGS Effect Sizes (per standard deviation change) and any additional relevant information regarding PGS performance were not reported.

Evaluated Samples

The evaluation phase relied on a single sample set, identified as **PSS012090**.

- **Total Sample Size:** 12,677 individuals.
- **Phenotype Definitions and Methods:** Defined as "Atrial fibrillation or flutter."
- **Sample Ancestry:** Exclusively European.
- **Cohort:** MHI.
- **Missing Data:** Information regarding participant follow-up time, age of study participants, additional ancestry descriptions, and other cohort/sample information was not reported.

DATASET NAME : PGS005296

PGS005297. The score was officially released in the PGS Catalog on **December 18, 2025**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term [MONDO_0005010](#).

Score Details and Construction

The formal name of this Polygenic Score is **Jorda_CAD_SingleTrait-PGS_females_2024**. It was constructed utilizing the **LDpred2** development method. The specific parameters configured for this development method were a Grid method with ρ (rho) set to 0.0018, h2SNP set to 0.0207, and a sparsity of 0.

The score was built upon the original genome build **GRCh38** and consists of exactly **1,042,395 genetic variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID [PGP000759](#). This score originates from the publication cited as [Jordà P et al. NPJ Genom Med \(2025\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **63,731 individuals**, representing 100% of the dataset. The ancestral composition for this phase was classified entirely (100%) as a Multi-ancestry group (including European populations).
- **Score Development/Training:** This phase utilized a pool of **997 individuals**, drawn entirely (100%) from populations of European ancestry.
- **PGS Evaluation:** The evaluation phase was conducted across **1 Sample Set**, with 100% of the participants classified as having European ancestry.

Development Samples

The development phase relied on two distinct sample sets—one for the initial GWAS and one for score development/training.

1. Source of Variant Associations (GWAS) The Genome-Wide Association Study (GWAS) dataset encompasses **63,731 individuals**. The sample ancestry is listed as European and Not Reported (NR). The specific cohort was also Not Reported (NR). This data was sourced from study identifiers in the GWAS Catalog ([GCST004787](#)) and Europe PMC ([28714975](#)).

2. Score Development/Training The training dataset encompasses **997 individuals** of exclusively European ancestry, sourced from the **MHI** cohort. The study identifier for this dataset is found in Europe PMC under the DOI [10.1038/s41525-025-00515-3](#). The phenotype definitions and methods for this group were based on "Atrial fibrillation or flutter." Details regarding the age of study participants, participant follow-up time, additional ancestry descriptions, and additional sample/cohort information were not reported.

Performance Metrics

Note: The PGS Catalog includes a disclaimer that performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated under a single metric ID, **PPM023005**, sourced from the Jordà P *et al.* NPJ Genom Med (2025) publication ([PGP000759](#)).

The model was evaluated for the reported trait of "Prevalent coronary artery disease defined as myocardial infarction, percutaneous transluminal coronary angioplasty, coronary artery bypass grafting, unstable angina and chronic ischemic heart disease." The statistical covariates included in the model were age, sex, and the four first principal components of genetic ancestry.

The evaluation yielded a classification metric **AUROC of 0.68603** and an **R² (Other Metric) of 0.10575**. The PGS Effect Sizes (per standard deviation change) and any additional relevant information regarding PGS performance were not reported.

Evaluated Samples

The evaluation phase relied on a single sample set, identified as [PSS012091](#).

- **Total Sample Size:** 5,152 individuals.
- **Sex Distribution:** The sample set is composed of **0.0 % Male samples** (indicating an entirely female cohort, which aligns with the "females" designation in the PGS name).
- **Phenotype Definitions and Methods:** Defined as "Atrial fibrillation or flutter."
- **Sample Ancestry:** Exclusively European.
- **Cohort:** MHI.
- **Missing Data:** Information regarding participant follow-up time, age of study participants, additional ancestry descriptions, and other cohort/sample information was not reported.

DATASET NAME: PGS005298

PGS005298. The score was officially released in the PGS Catalog on **December 18, 2025**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term **MONDO_0005010**.

Score Details and Construction

The formal name of this Polygenic Score is **Jorda_CAD_SingleTrait-PGS_males_2024**, indicating a male-specific focus in its development or application. It was constructed utilizing the **LDpred2** development method. The specific parameters configured for this development method were a Grid method with ρ (rho) set to 0.0056, h2SNP set to 0.0207, and a sparsity of 0.

The score was built upon the original genome build **GRCh38** and consists of exactly **1,042,395 genetic variants**. The effect weight type assigned to these variants is categorized as **beta**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID **PGP000759**. This score originates from the publication cited as [Jordà P et al. NPJ Genom Med \(2025\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **63,731 individuals**, representing 100% of the dataset. The ancestral

composition for this phase was classified entirely (100%) as a Multi-ancestry group (including European populations).

- **Score Development/Training:** This phase utilized a pool of **1,503 individuals**, drawn entirely (100%) from populations of European ancestry.
- **PGS Evaluation:** The evaluation phase was conducted across **1 Sample Set**, with 100% of the participants classified as having European ancestry.

Development Samples

The development phase relied on two distinct sample sets—one for the initial GWAS and one for score development/training.

1. Source of Variant Associations (GWAS) The Genome-Wide Association Study (GWAS) dataset encompasses **63,731 individuals**. The sample ancestry is listed as European and Not Reported (NR). The specific cohort was also Not Reported (NR). This data was sourced from study identifiers in the GWAS Catalog ([GCST004787](#)) and Europe PMC ([28714975](#)).

2. Score Development/Training The training dataset encompasses **1,503 individuals** of exclusively European ancestry, sourced from the **MHI** cohort. The study identifier for this dataset is found in Europe PMC under the DOI [10.1038/s41525-025-00515-4](#). The phenotype definitions and methods for this group were based on "Atrial fibrillation or flutter." Details regarding the age of study participants, participant follow-up time, additional ancestry descriptions, and additional sample/cohort information were not reported.

Performance Metrics

Note: The PGS Catalog includes a disclaimer that performance metrics are displayed exactly as reported by the source studies and are not necessarily comparable with one another due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated under a single metric ID, **PPM023006**, sourced from the Jordà P *et al.* NPJ Genom Med (2025) publication ([PGP000759](#)).

The model was evaluated for the reported trait of "Prevalent coronary artery disease defined as myocardial infarction, percutaneous transluminal coronary angioplasty, coronary artery bypass grafting, unstable angina and chronic ischemic heart disease." The statistical covariates included in the model were age, sex, and the four first principal components of genetic ancestry.

The evaluation yielded a classification metric **AUROC of 0.69191** and an **R² (Other Metric) of 0.15574**. The PGS Effect Sizes (per standard deviation change) and any additional relevant information regarding PGS performance were not reported.

Evaluated Samples

The evaluation phase relied on a single sample set, identified as [PSS012092](#).

- **Total Sample Size:** 7,525 individuals.
- **Phenotype Definitions and Methods:** Defined as "Atrial fibrillation or flutter."

- **Sample Ancestry:** Exclusively European.
- **Cohort:** MHI.
- **Missing Data:** Information regarding participant follow-up time, age of study participants, additional ancestry descriptions, and other cohort/sample information was not reported.

CURRENT DATASET BEING USED : PGS002809

PGS002809. The score was officially released in the PGS Catalog on **November 7, 2022**. The reported predicted trait for this score is "Coronary artery disease." This is formally mapped to the trait "[coronary artery disorder](#)" in the ontology, identified by the term [MONDO_0005010](#).

Score Details and Construction

The formal name of this Polygenic Score is **GRS_CAD**. It was constructed utilizing the **Genome-wide significant SNPs** development method, though specific parameters for this method were Not Reported (NR).

The original genome build for this score was Not Reported (NR). The score consists of exactly **205 genetic variants**. The effect weight type assigned to these variants is categorized as **log(OR)**.

The source of this Polygenic Score within the PGS Catalog is registered under the Publication ID [PGP000388](#). This score originates from the publication cited as [Ahmed R et al. Int J Cardiol Heart Vasc \(2022\)](#).

Ancestry Distribution Summary

The ancestry representation across the different phases of this score's development and evaluation is detailed below:

- **Source of Variant Associations (GWAS):** This initial phase involved an aggregate pool of **2,202,086 individuals**, representing 100% of the dataset. The ancestral composition for this phase breaks down as 38.6% Multi-ancestry (including European), 31.3% European, 20.2% Not Reported, and 10% East Asian populations.
- **PGS Evaluation:** The evaluation phase was conducted across **1 Sample Set**, with 100% of the participants classified as having European ancestry.

Development Samples: Source of Variant Associations (GWAS)

The Genome-Wide Association Study (GWAS) dataset encompasses 2,202,086 individuals drawn from 11 distinct study subsets.

Subsets without designated GWAS Catalog IDs:

- **Europe PMC [23202125](#)**: Contributed 194,427 individuals of European and South Asian ancestry. Cohort information was Not Reported (NR).
- **Europe PMC [28209224](#)**: Contributed 120,575 individuals of European ancestry. Cohort information was Not Reported (NR).
- **Europe PMC [28530674](#)**: Contributed 250,736 individuals spanning European, South Asian, East Asian, and African American or Afro-Caribbean ancestries. Cohort information was Not Reported (NR).
- **Europe PMC [28584231](#)**: Contributed 147,798 individuals whose ancestry and cohort were both Not Reported (NR).

Subsets with designated GWAS Catalog IDs:

- **[GCST000998](#) / Europe PMC [21378990](#)**: Contributed 86,995 individuals of European ancestry spanning 12 distinct cohorts.
- **[GCST004787](#) / Europe PMC [28714975](#)**: Contributed 63,731 individuals with European and Not Reported (NR) ancestries. The specific cohort was Not Reported (NR).
- **[GCST005194](#) / Europe PMC [29212778](#)**: Contributed 296,525 individuals with Not Reported (NR) ancestry, drawn from the UKB cohort.
- **[GCST010479](#) / Europe PMC [32469254](#) (Subset 1)**: Contributed 51,442 individuals of East Asian ancestry. Cohort information was Not Reported (NR).
- **[GCST010479](#) / Europe PMC [32469254](#) (Subset 2)**: Contributed 340,799 individuals of East Asian and European ancestries. Cohort information was Not Reported (NR).
- **[GCST010866](#) / Europe PMC [33020668](#) (Subset 1)**: Contributed 480,830 individuals of European ancestry. Cohort information was Not Reported (NR).
- **[GCST010866](#) / Europe PMC [33020668](#) (Subset 2)**: Contributed 168,228 individuals of East Asian ancestry. Cohort information was Not Reported (NR).

Performance Metrics

Disclaimer: The PGS Catalog notes that performance metrics are displayed exactly as reported by the source studies. They are not necessarily comparable across studies due to differences in sample characteristics, phenotyping, and statistical modeling.

The performance of the score was evaluated under two distinct metric IDs sourced from the [Ahmed R et al. Int J Cardiol Heart Vasc \(2022\)](#) publication ([PGP000388](#)). Both metrics were evaluated on the [PSS009989](#) sample set, which comprises 360,098 individuals of European ancestry. Standard classification metrics, PGS effect sizes, and covariates included in the models were not reported for either metric.

- **Metric PPM015577**: Evaluated for the reported trait of "Incident coronary artery disease." The evaluation produced a Hazard Ratio (HR, ≥ 3 vs < 0.5 risk) of **3.02** with a confidence interval of **[2.73, 3.33]**. This metric was calculated as a Population-standardized GRS.
- **Metric PPM015578**: Evaluated for the reported trait of "Incident coronary artery disease in subjects with borderline-/intermediate-ASCVD risk." The evaluation produced a Hazard Ratio (HR, ≥ 3 vs < 0.5 risk) of **2.91** with a confidence interval of **[2.59, 3.26]**. No additional relevant PGS performance information was reported.

Evaluated Samples

The evaluation phase relied on a single sample set, identified as [PSS009989](#).

- **Total Sample Size:** 360,098 individuals.
- **Sex Distribution:** 45.1% Male samples.
- **Participant Age:** Mean age of 56.95 years.
- **Sample Ancestry:** Exclusively European.
- **Cohort:** UKB.
- **Missing Data:** Information regarding participant follow-up time, phenotype definitions and methods, and additional sample/cohort descriptions was not reported.